

Characteristics of Forces — Paper B

MEDIUM • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**, and every resultant force must carry a **direction**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: **weight = mass × gravitational field strength** ($W = mg$), and on Earth **$g = 10 \text{ N/kg}$** .

SECTION 1 – DRAWING AND MEASURING FORCES

- 1.** A crate rests on the floor. In the space below, draw the crate and label the **two** vertical forces acting on it, using arrows. Then state what you notice about the lengths of your two arrows, and why. [4]

- 2.** Explain why a force is drawn as an **arrow** on a diagram, rather than written as a number on its own. [3]

- 3.** A newtonmeter is used to measure a force.
(a) State the unit that a newtonmeter reads in. [1]

(b) Name the part inside a newtonmeter that stretches. [1]

(c) State what happens to the stretch when the force pulling on it is doubled. [1]

SECTION 2 – RESULTANTS AND MOTION

4. Calculate the resultant force in each case. Give a direction as well as a size.

(a) 80 N to the right and 50 N to the right. [1]

(b) 90 N to the left and 35 N to the right. [1]

(c) 120 N to the right and 120 N to the left. [1]

(d) 6 N to the right, 14 N to the left and 4 N to the right. [2]

5. A lorry drives along a straight motorway at a steady 25 m/s. The driving force from its engine is 4500 N.

(a) State the size of the total backward force acting on the lorry. [1]

(b) Explain how you know. [1]

- (c) The driver presses the accelerator and the driving force rises to 5200 N. [2]
Calculate the new resultant force and describe what happens to the lorry.

6. Explain the difference between **balanced** and **unbalanced** forces, and state what [3]
each does to an object that is already moving.

SECTION 3 – FRICTION

7. Friction acts on a moving object.

- (a) State the direction in which friction acts. [1]

- (b) Give two everyday situations in which friction is useful. [2]

- (c) Give one situation in which friction is a nuisance. [1]

8. A cyclist stops pedalling on a flat road and gradually comes to a stop.

- (a) Name the two forces that slow her down. [2]

- (b) Explain, in terms of the resultant force on her, why she slows down. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.